Ayush Pratap Singh

+91 7889440379 ayushtomar8624@gmail.com LinkedIn Github

Summary

Software Engineer with 1 year of experience building backend, data engineering, and event-driven systems for enterprise risk and compliance platforms.

Education

Chitkara University, Chandigarh **2022 - 2026**
B.E. in Computer Science & Engineering *CGPA: 9.32/10*

Technical Skills

Frontend: JavaScript, TypeScript, React.js (Hooks, Redux, TanStack Query), Next.js (SSR, App Router)
Backend & Data: Node.js, Express.js, Python, Java, FastAPI, Kafka, Apache Airflow, ETL Pipelines, SQL, PostgreSQL, MongoDB, Neo4j
AI & LLM: LangChain, LangGraph, RAG, Prompt Engineering, Vector Search
Cloud, DevOps & Security: AWS (S3, EC2), Docker, CI/CD (Jenkins), Git, Linux (RHEL/Debian), Authentication, OIDC, Kerberos
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), OS, Computer Networks

Experience

Morgan Stanley **Aug 2025 – Present**
Software Developer (Apprentice) *Bengaluru, India*

- Refactored and extended an event-driven policy-as-code compliance platform from POC to MVP by introducing **Pydantic** validation, refining inter-service **Kafka** message contracts, hardening failure-handling paths, and implementing automated stakeholder notifications for compliance scan failures.
- Architected an end-to-end regression validation framework that verified Airflow workflow execution, S3 data integrity, Neo4j graph updates, relationship consistency, and incremental-load reconciliation for created, updated, and deleted entities, reducing release validation time from 2–3 days to 5–7 hours through automated execution and reporting.
- Designed and implemented a multi-agent RAG assistant using LangGraph, featuring LLM-based intent detection, supervisor-driven orchestration, task-specific **ReAct** agents for risk-score retrieval and documentation search, and a reflection stage for response quality evaluation before final answer generation.
- Implemented **Apache Airflow** ETL workflows to onboard new risk data sources into an enterprise fitness-scoring platform, handling source-specific ingestion, transformation, and scoring requirements while introducing incremental (delta) processing that reduced runtime by 30–40%.
- Optimized large-scale Neo4j inserts by rewriting multi-transaction logic with **UNWIND**, significantly reducing database round trips and ingestion time.

Projects

Outreach | *Next.js, TypeScript, MongoDB, Gemini AI, Cheerio* **GitHub** | **Live**

- Built and Deployed an end-to-end cold email pipeline that integrates LLM-based content generation with automated web-scraping to deliver highly relevant, company-specific outreach.
- Implemented Cheerio-based scraping logic to extract real-time organizational data, which is parsed and passed to Gemini AI for context-aware email drafting.
- Engineered a CRUD-based subscriber management system with segmentation, CSV parsing, and authentication, providing a secure backend for campaign lifecycle management.

Flex It Out | *Next.js, TypeScript, TensorFlow.js, Node.js* **GitHub** | **Live**

- Built and deployed a fitness platform that helps users track workouts, complete challenges, and maintain exercise routines through gamified engagement.
- Integrated TensorFlow.js-based pose detection to analyze exercise movements in real time and provide instant feedback on workout execution.
- Implemented a reward system with daily challenges, achievement tracking, and company-sponsored incentives to encourage consistent user participation.